Kinematics and Dynamics of Machines

Assignment-5

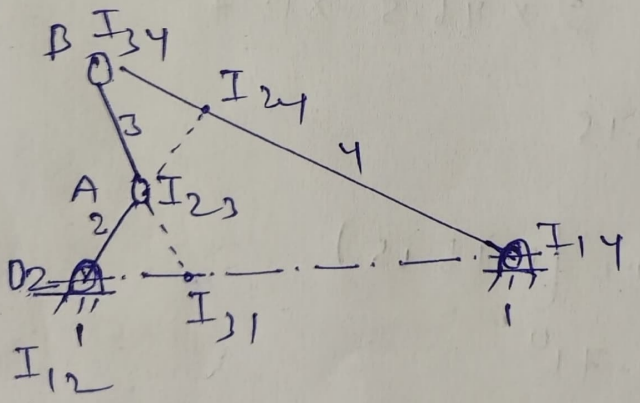
Pushup Kinematics

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Suitable position: When legs hips and back are straight in a line.

Comfortable duration: At least 1 minute

Considering the linkages with link lengths;



$r_1 = 127$

$r_2 = 23.5$

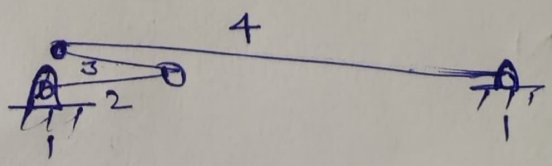
$r_3 = 23$

$r_4 = 128$

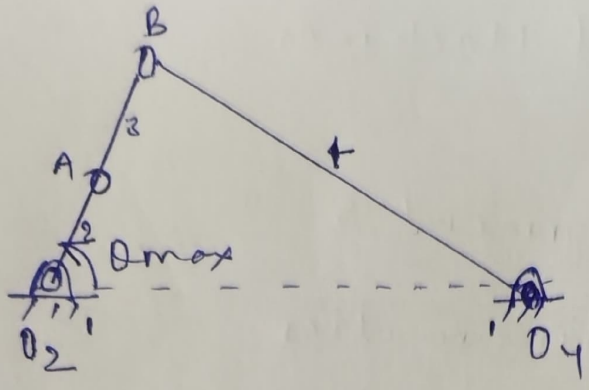
In the above diagrams instantaneous centres are shown.

Limiting positions:-

(1)



(2)



This limiting position is because rotation beyond this is not possible in anticlockwise direction because elbow joint (or joint A) wouldn't allow it.

In $\Delta O_2 B O_4$

$$\cos\theta_{max} = \frac{46.5^2 + 127^2 - 128^2}{2 \times 46.5 \times 127}$$

$$= 0.1615$$

$$\Rightarrow \theta_{max} = \cos^{-1}(0.1615)$$

$$= 80.71^\circ$$

Now, weight acting on centre of mass G

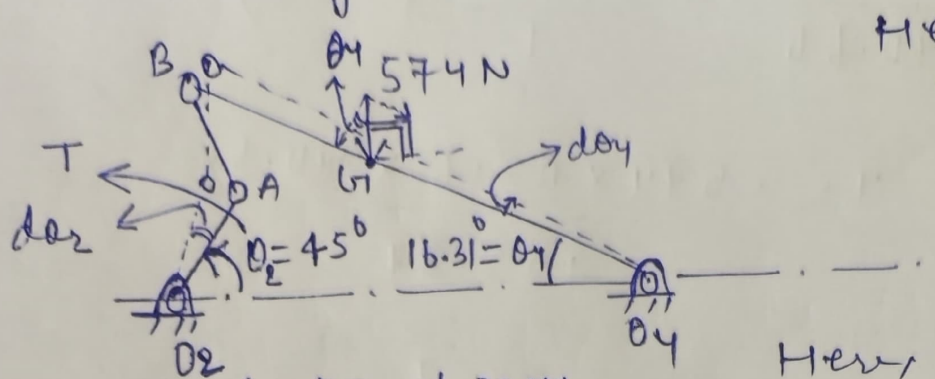
$$W = mg$$

$$= 58.5 \text{ kg} \times 9.81 \text{ m/s}^2$$

$$= 573.88 \text{ N}$$

$$\Rightarrow W \approx 574 \text{ N}$$

Now at a posture shown in figure;



Here moment T is acting on elbow joint

G1: Centre of mass

Here, $4O_4 = 98 \angle 163.69^\circ$

Using Method of virtual work,

$$dU = 0$$

$$\Rightarrow T\hat{i} \cdot d\theta_2 \hat{i} + F(-\hat{j})\left(-R\,d\theta_4(\cos\theta_4 \hat{j} + \sin\theta_4 \hat{i})\right) = 0$$

$$\Rightarrow T\,d\theta_2 + F\cos\theta_4 \times \frac{98}{100} d\theta_4 = 0$$

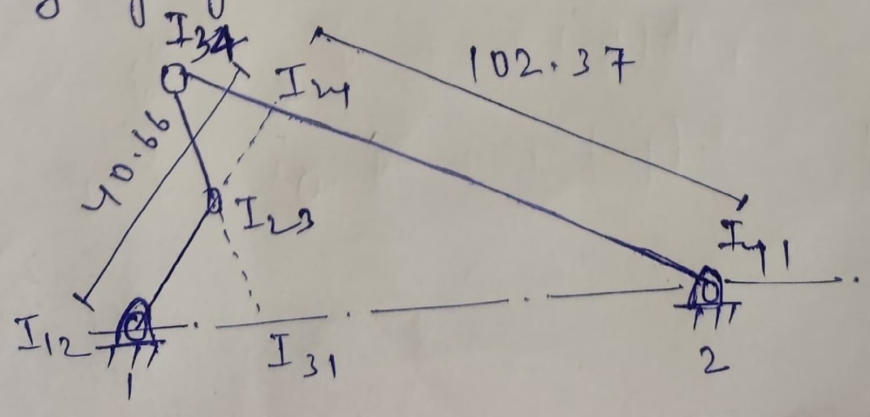
$$\Rightarrow T\,d\theta_2 = -F\cos\theta_4\ 0.98\ d\theta_4$$

$$\Rightarrow \frac{d\theta_2}{d\theta_4} = \frac{-F}{T}\cos 16.31 \times 0.98$$

$$\Rightarrow \frac{\omega_2}{\omega_4} = \frac{-F}{T}(0.96)$$

Using formula, $\frac{\omega_2}{\omega_4} = \frac{I_{41}\, I_{24}}{I_{21}\, I_{24}}$

Using graphical analysis;



$$\Rightarrow \frac{102.37}{40.66} = \frac{-574}{T}\ (0.96)$$

$$\rightarrow T = \frac{-574 \times 0.96 \times 40.66}{102.37}$$

$$\Rightarrow \boxed{T = -218.86\ Nm}$$

$\therefore$ Torque, $T = 218.86\,(-\hat{k})$ CW

Yes, pushups can improve cardiac health. They can help strengthen the heart muscles, improve circulation of blood, and reduce the risk of cardiovascular diseases. However, it is important to practice the right way of performing pushups, to avoid injuries.

Method adopted:-

We have used the method of virtual work to solve this pushup kinematics problem.